

Extracting Information from Big Data: Issues of Measurement, Inference and Linkage

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Introduction¹

Big Data pose several interesting and new challenges to statisticians and others who want to extract information from data. As Robert Groves pointedly commented, the era is “appropriately called Big Data and not Big Information”², because there is a lot of work for analysts before information can be gained from “auxiliary traces of some process that is going on in society.” The analytic challenges most often discussed are those related to three of the V’s that are used to characterize Big Data³. The *volume* of truly massive data requires expansion of processing techniques that match modern hardware infrastructure, cloud computing with appropriate optimization mechanisms and re-engineering of storage systems⁴. The *velocity* of the data calls for algorithms that allow learning and updating on a continuous basis, and of course the computing infrastructure to do so. Finally, the *variety* of the data structure requires statistical methods that more easily allow for the combination of different data types collected at different levels, sometimes with a temporal and geographic structure.

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² Robert Groves talk at the World Bank on December 19th 2012 event “What Happens When Big Data Meets Official Statistics” <http://live.worldbank.org/what-happens-when-big-data-meets-official-statistics-live-webcast> [last accessed 01/20/2014]

³ O’Neil and Shutt 2013, Doing Data Science .. O’Reilly Media

⁴ Agrawal, D., Bernstein, P., Bertino, E., Davidson, S., Dayal, U., Franklin, M., ... & Widom, J. (2012). Challenges and Opportunities with Big Data. A community white paper developed by leading researchers across the United States.

However when it comes to *privacy* and *confidentiality*, the challenges of extracting (meaningful) information from Big Data are in our view quite similar to those associated with data of much smaller size, surveys being one example. For statisticians and any quantitative working (social) scientist there are two main concerns when it comes to extracting information from data, that we summarize here as concerns about *measurement* and concerns about *inference*. Both of these aspects can be implicated by privacy and confidentiality concerns.

- (a) By questions of *measurement* we mean: Are the right key variables and all covariates in the data to answer the research question? Are there any unobserved variables that confound the analysis when uncontrolled for? Is there systematic measurement error in the measures? What is the best unit of analysis on which the measures are taken?
- (b) By questions of *inference* we mean: Do we understand the sampling process? Are all units we need in the analysis? Are certain units systematically missing? Do some units appear multiple times? Do we have all measures on all units that we need? Whom do the units represent?

In settings where data collections (big or small) are specifically *designed* to answer specific scientific research questions, the answers to the questions above are (usually) relatively easy to determine or, better yet, are taken into account when the data collected. But as soon as data are used for purposes other than those they were designed for, or when the *data-generating process* breaks down – as is unfortunately often the case, both with data collected for experimental purposes and with data from sample surveys and Censuses-- this is no longer the case. Neither is it the case for most Big Data sources.

There are many reasons for breakdowns in the data- generating process of traditional data as well as Big Data; *actual and perceived* privacy and confidentiality threats can be one of them, though usually more on the measurement side than on the inference side. This might be very different with Big Data. Irrespective of the size of the data at hand, researchers have to understand the data-generating process in order to determine if statistical model assumptions are met and to know if all relevant information is indeed included in the data for a given research question. Here is where

we see a key difference between data commonly referred to as Big Data and more traditional data sources.

For traditional data, methodology has been developed to overcome problems in the data-generating process. Among survey methodologists, a guiding principle for detecting problems is the Total Survey Error framework⁵, and statistical methods for weighting, calibration and other forms of adjustments⁶ are commonly used to mitigate errors in the survey process. Likewise for ‘broken’ experimental data, techniques like propensity score adjustment and principal stratification are widely used to fix flaws in the data-generating process⁷. Some of this methodology is likely transferable to Big Data, but some of it might need to be tweaked. Most likely solutions will lie in a combination of traditionally designed data and Big Data. However, such solutions of *data linkage and information integration* are themselves threatened by concerns about privacy and confidentiality.

Data-Generating Processes

When we look closely, it is the data-generating process that differentiates Big Data from more traditional types of data, such as Censuses, sample surveys, and experimental data. Even with those more traditional data, we would, at least once in a while, see data of large volume (e.g. the U.S. Census), data of high velocity (e.g. Nielsen TV meter⁸), and data with a wide variety of different variables and a complex structure (e.g. National Health and Nutrition Examination Survey⁹). What is different is the data collection for all three of these examples was specifically *designed* with a research question in mind.

In the case of designed data collection, the research questions are translated into concepts that need to be measured (e.g. employment), measurement instruments are developed to best capture these

⁵ Groves, R.M. and Lyberg, L. (2010). Total Survey Error. *Public Opinion Quarterly*, 74, 5, 849-879.

⁶ For an overview see Valliant, Dever, Kreuter 2013: *Practical Tools for Sampling and Weighting*

⁷ Rosenbaum, R. and Rubin, D.B. (1983) The Central Role of the Propensity Score in Observational Studies for Causal Effects. *Biometrika*, Vol. 70, No. 1. (Apr., 1983), pp. 41-55; Frangakis, C. and Rubin, D. (2001), Principal stratification in causal inference. In *Biometrics*, 2002, 58, 21–29.

⁸ <http://www.nielsen.com/us/en/nielsen-solutions/nielsen-measurement/nielsen-tv-measurement.html>

⁹ <http://www.cdc.gov/nchs/nhanes.htm>

concepts (e.g. people that did any work for pay or profit during the survey week, or those with a job but not working during the survey week because of illness, vacation, leave etc.¹⁰), populations are defined for which measures should be taken (e.g. U.S. Civilian noninstitutionalized population 16 years and older¹¹), and (usually) samples are drawn to make data collections more efficient. If used for official statistics these design features are transparent and made public. For other data products the Transparency Initiative of the American Association for Public Opinion Research¹² now has guidelines in place to match such transparency. A slightly different approach is taken by ESOMAR but also with the goal of creating transparency for the data-generating process¹³.

It is this transparency that allows an evaluation of the quality and usefulness of the data for a given research question--for example, knowledge about people that were sampled to be in the survey but could not be reached, or decided not to participate. If the error mechanism is known, researchers can evaluate potential bias. Statisticians express such bias, for example for an estimated mean, as a function of the rate of response and the difference between respondents and nonrespondents or, if response is seen as stochastic process, as $\text{Bias}(\bar{y}_r) = \frac{\sigma_{yp}}{\bar{p}}$ where σ_{yp} is the covariance between y and the propensity to (in this case) respond to the survey, ρ , and $\bar{p}$ is the mean propensity over the sample. Should, for example, sampled cases with privacy concerns be less likely to participate in a survey¹⁴ but those privacy reasons are unrelated to the topic of the survey, then no bias will appear due to (this) breakdown in the process¹⁵.

In contrast, the data-generating process of many data streams is less transparent and much less under the control of the researcher. Many Big Data streams arise through systems that create data as a by-product of human behavior, usually in interaction with a computer. Credit card transactions, loyalty cards, Twitter posts, Facebook entries, cell phone usage, all those behaviors

¹⁰ http://www.bls.gov/cps/cps_hgtm.pdf [last accessed 01/20/2014]

¹¹ http://www.bls.gov/cps/cps_over.htm#coverage [last accessed 01/20/2014]

¹² http://www.aapor.org/Transparency_Initiative.htm#.Ut2pHrQo7IU [last accessed 01/20/2014]

¹³ <http://www.esomar.org/news-and-multimedia/news.php?idnews=104>

¹⁴ For studies that show this effect see Singer, E. (2004). Confidentiality, risk perception, and survey participation. *Chance*, 17(3), 30-34. And Singer, E., Mathiowetz, N., and Couper, M.P. 1993. The Role of Privacy and Confidentiality as Factors in Response to the 1990 Census. *Public Opinion Quarterly*, 57, 465-82.

¹⁵ For a meta-analysis on nonresponse bias see Groves and Peytcheva XXX.

leave electronic traces that arise *organically*¹⁶. It is this difference between *designed* and *organic* data that poses extra challenges to the extraction of information.

Note, some Big Data share the features of designed data, such as environmental monitors which are built to measure a specific types of pollution. Their locations are carefully selected and their measurement times are either selected by design or continuous, removing any ‘organic’ element from the data-generating process. However, even in this example, the data collected by environmental monitoring agencies are often used for purposes for which they were not designed. The most common example perhaps is estimation of the health impacts of ambient air pollution.

The use of data for other than the designed purpose is also reflected in Horrigan’s¹⁷ comment, who characterizes Big Data as “nonsampled data, characterized by the creation of databases from electronic sources whose primary purpose is something other than statistical inference.” In designed data (ideally) all measures taken have to fulfill a purpose with respect to the research question (and a related statistic of interest), whereas with Big Data, the data fulfill a purpose for a particular action (i.e. the credit card payment, the reception of a phone call, or the expression of ‘liking’), but not for a particular research question unless the research question is a mere measure of the number of actions of that particular type.

The use of organic data for other than their immediate purpose within the system in which they arise is not peculiar to the Big Data area. For decades researchers have used administrative data in statistical analyses. Administrative data are the result of government or other administrative processes; prime examples of data often increasingly used for statistical analyses are tax data¹⁸, or data produced through social security notifications¹⁹. The variables that are contained in those data are usually those necessary for administering a program or administrative task, and measure what

¹⁶ Groves, R. M. (2011). Three eras of survey research. *Public Opinion Quarterly*, 75(5), 861-871.

¹⁷ <http://magazine.amstat.org/blog/2013/01/01/sci-policy-jan2013/> [last accessed 01/20/2014]

¹⁸ For a recent example see: Equality of Opportunity project (Ray Chetty, Nathaniel Hendren, Patrick Kline, Emmanuel Saez and colleagues) using Earned Income Tax Credit to study the impact of tax expenditures on intergenerational mobility. <http://www.equality-of-opportunity.org/>.

¹⁹ Schmieder, Johannes; Wachter, Till von; Bender, Stefan (2012): The effects of extended unemployment insurance over the business cycle * evidence from regression discontinuity estimates over 20 years. In: The Quarterly Journal of Economics, Vol. 127, No. 2, S. 701-752.

is necessary to evaluate program-relevant features. When these data are used to extra information that go beyond the description of the program, researchers often quickly realize that the measures do not match up with the concepts they try to measure, and that often not all cases of interest are captured through these administrative data. For example in the social security notifications that form the basis of the administrative data used in a study of workplace heterogeneity and wage inequality with German administrative data²⁰, education is known to have many missing values or conflicting data within a person across employment spells, and certain groups like civil servants and the self-employed are not included in the data base at all.

Another mechanism by which Big Data often arise is in the merging of multiple databases, studies, surveys, or datasets. Computing power today has allowed “mega-databases” to be formed to address new and interesting questions using data in novel ways. However, two key concerns when merging takes place is that (1) now multiple datasets are being used for purposes for which they were not designed; and (2) data that once retained a subject’s privacy, may no longer do so when they are merged with another database.

Challenge: Measurement

There is much to say about possible measurement problems when using Big Data, for example challenges in text mining of Twitter or Facebook feeds to distinguish between sarcasm, irony, humor, actions vs. intentions etc. Some recent examples showed promise in solutions to automate such “feature selection”²¹. Here we focus on those measurement problems that are likely induced by privacy, confidentiality and related issues: (1) Misreporting in anticipation of perceived data use and (2) lack of model-relevant variables.

We know from survey research that people misreport or underreport socially undesirable behavior and illegal activities to avoid unfavorable impressions and legal consequences should the information be disclosed and overreport desirable behavior—e.g. voting²². Preliminary research

²⁰ Card, David; Heining, Jörg; Kline, Patrick (2013): Workplace heterogeneity and the rise of West German wage inequality. In: The Quarterly Journal of Economics, Vol. 128, No. 3, pp. 967-1015.

²¹ For a discussion and examples of feature selection see <http://web.eecs.umich.edu/~michjc/papers/ringtail.pdf>

²² Tourangeau, R., and T. Yan. 2007. "Sensitive questions in surveys." *Psychological Bulletin*, 133(5): 859-883.

reports reasonable accurate reflection of user characteristics on social network sites²³, though honesty is suspected to vary across personality types²⁴. Not much is known on how the report of attitudes and behaviors varies across networks, topics and (perceived) use of the data and the associated *actual and perceived privacy* issues. In survey data collections self-administered modes have been shown to mitigate effects of social desirability compared to interviewer administered modes²⁵. But even if the mode is similar to those used in self-administered surveys (Web and text-messages), the difference here is that the primary purpose of generating these data includes an intended readership. Also it is possible that what is shared is of good quality, but what matters most is not shared. Likewise a *privacy* related observation issue can be the sharing/posting/reporting of information from people other than oneself. This trade-off between observation and non-observation error in Big Data deserves attention. It would be a sad state if increased research use of social media or other Big Data sources changes the very nature of these sites.

A second feature of many Big Data sources that can easily create measurement problems is the fact that while they are ‘case rich’ they tend to be ‘variable poor.’²⁶ The Billion Prices Project²⁷, for example can give timely and detailed information if the purpose is to only measure prices of consumer goods, but as soon as the research interest shifts to understanding how price changes lead to shifts in household expenditures a single source of Big Data will no longer suffice²⁸. To perform household level analysis consumption and purchase data need to be linked up at a household level, which might be doable if confidential credit card billing information from websites like Amazon, Paypal etc. are shared, though this obviously would require a data infrastructure that is currently not in place (see suggestions Wilbanks in this volume).

²³ Brown, V. R., & Vaughn, E. D. (2011). The writing on the (Facebook) wall: The use of social networking sites in hiring decisions. *Journal of Business and Psychology*, 26(2), 219-225.

²⁴ Karl, K., Peluchette, J., & Schlaegel, C. (2010). Who's Posting Facebook Faux Pas? A Cross-Cultural Examination of Personality Differences. *International Journal of Selection and Assessment*, 18(2), 174-186

²⁵ F Kreuter, S Presser, R Tourangeau (2008) [Social Desirability Bias in CATI, IVR, and Web Surveys The Effects of Mode and Question Sensitivity](#) *Public Opinion Quarterly* 72 (5), 847-865

²⁶ A point made by Couper, M. P. (2013, December). Is the Sky Falling? New Technology, Changing Media, and the Future of Surveys. In *Survey Research Methods* (Vol. 7, No. 3, pp. 145-156) and Prewitt, K. (2013). The 2012 Morris Hansen lecture: Thank you Morris, et al., for Westat, et al. *Journal of Official Statistics*, 29(2), 223-231. This is different from the biomedical sciences where often few cases have countless variables.

²⁷ Bpp.mit.edu

²⁸ Horrigan, M. 2013. Big Data: A perspective from the BLS. *AMSTATNews*. 1. January, 2013.

A recent project at the University of Michigan²⁹ uses Twitter data to predict Initial Claims for Unemployment Insurance using the University of Michigan Social Media Job Loss Index. The prediction is based on a factor analysis of social media messages mentioning job loss and related outcomes³⁰. As part of this project, auxiliary information was linked to Twitter messages via geographic location information, a powerful tool but not without its *own threats to privacy*. O’Neil and Shutt also point to the usefulness of location data for the correct interpretation of timestamp data to know the time zone users are in.³¹ Obtaining such information and other additional covariates can also be used to make sense of outliers that would otherwise be removed from analyses, a common strategies in dealing with these types of data³².

Challenge: Inference

One interesting feature of Big Data is that because there is so much data, it is easy to overlook what is missing. At first glance one might think that there is no problem of inference with Big Data. After all, with a constant stream of data and the technological capacity to capture all the data, some state that sampling is no longer needed but the population can be analyzed instead³³. But what is the population? If Amazon is interested in the correlation between the purchases of two different books on amazon.com, they can of course analyze the population and no inference problem would appear. But if for example Twitter is used to measure health or political attitudes the problem of inference is much harder, and starts with a specification of the unit of analysis. Paul and Dredze³⁴ reported good success with population level metrics in their analysis of public health,

²⁹ <http://econprediction.eecs.umich.edu/>

³⁰ Antenucci, D., M. Cafarella, M. Levenstein, and M. Shapiro. (2012). *Creating Measures of Labor Market Flows Using Social Media*. Presentation to the National Bureau of Economic Research, Cambridge, MA, July 16

³¹ O’Neil and Shutt 2013, *Doing Data Science*. O’Reilly Media

³² Yan and Olson (2013). *Analyzing Paradata to Investigate Measurement Error*. In Kreuter, F. (2013). *Improving Surveys with Paradata: Making use of Process Information*. wiley

³³ Mayer-Schönberger, V. and Cukier, K, (2013). *Big Data: A Revolution That Will Transform How We Live, Work and Think*. John Murray.

³⁴ Paul, M. and Dredze, M. *You Are What You Tweet: Analyzing Twitter for Public Health*. Michael J. Paul and Mark Dredze. Human Language Technology Center of Excellence.
www.cs.jhu.edu/~mdredze/publications/twitter_health_icwsm_11.pdf

but are hampered by the irregularity of postings when trying to answer questions on frequency of health problems, re-occurrence of health problems and the like.

As with any data source, researchers have to again ask themselves if the data-generating process is appropriate for the research question and captures the population to which they are trying to make inference. In statistical terms, the challenge is to assess undercoverage, overcoverage, multiplicity, as well as other forms of missing data and (here) their relationship to privacy and confidentiality.

Undercoverage

Regardless of whether one captures all available data (at a given time point) or just a sample of the data, the data set from which one draws cases (the frame) is almost never perfect. It is not hard to imagine that Big Data taken from the internet suffer from *undercoverage*, from the exclusion of units in the target population (those without access to internet). According to estimates from the Current Population Survey, the internet penetration rate in U.S. households was still below 80% in 2011³⁵, and almost 30% of individuals report no internet use anywhere (irrespective of access to a computer at home or elsewhere). Likewise, an estimated 30% of American adults do not have a credit card³⁶, which means again about a third of the population are not represented in data from commercial vendors.

The proportion of people with internet access that could be covered (if they would all use a particular service that is used for web scraping) is however only one piece and translates into *coverage bias* only when there is a sizable difference between those covered (e.g. by having internet access or a credit card) and those uncovered. Coverage bias can be expressed as follows³⁷:

³⁵ Thom. 2013. "Computer and Internet Use in the United States." Current Population Survey Reports, P20-568. U.S. Census Bureau, Washington, DC.

³⁶ <http://www.statisticbrain.com/credit-card-ownership-statistics/> find original source
<http://www.creditcards.com/credit-card-news/credit-card-industry-facts-personal-debt-statistics-1276.php>

³⁷ Lessler, J. T. and Kalsbeek, W. D. (1992). Nonsampling Error in Surveys. John Wiley and Sons.

$$\text{bias}(\bar{Y}_c) = \frac{N_{uc}}{N_{pop}} [\bar{Y}_c - \bar{Y}_{uc}] ,$$

with N_{uc} reflecting the number of people that are undercovered, N_{pop} the total number of people in the population, $\bar{Y}_c$ the mean for the covered units, and $\bar{Y}_{uc}$ the mean for the undercovered units. Unfortunately, as can be seen from this equation, while the rate of undercovered people to the population stays the same across all statistics derived from a given data source, the difference in a between covered and undercovered units can vary from variable to variable.

Several studies list potential indicators for bias when comparing those without internet access to those with access³⁸: In a German study persons without internet access are found to be less educated and slightly older than those with access³⁹. In a U.S. study, internet users are more likely to be in good health, to have health insurance, and to exercise regularly⁴⁰, and for the elderly in the U.S. significant differences between internet users and non-users on financial and health variables were detected⁴¹. If we look at specific Big Data sources both the undercoverage rate as well as the potential for undercoverage bias is even bigger; in 2013 only 13% of the U.S. online population actively tweets⁴² and nearly half of the Twitter users are under age 35 and only 2% are 65 or older⁴³.

Privacy and confidentiality concerns contribute to undercoverage, both at the person level but also at the level of individual postings, or transactions⁴⁴. Survey results from Google showed a strong

³⁸ For a review see Eckman, S. 2014. Did the Inclusion of Non-Internet Households in the LISS Panel Reduce Coverage Bias? Manuscript. Institute for Employment Research. Germany

³⁹ Bosnjak, M., Haas, I., Galesic, M., Kaczmirek, L., Bandilla, W., and Couper, M. P. (2013). Sample composition discrepancies in different stages of a probability-based online panel. *Field Methods*, 25(4), 339-360

⁴⁰ Dever, J. A., Rafferty, A., and Valliant, R. (2008). Internet surveys: Can statistical adjustment eliminate coverage bias? *Survey Research Methods*, 2(2), 47-62.

⁴¹ Couper, M.P., Kapteyn, A., Schonlau, M., and Winter, J. (2007). Noncoverage and nonresponse in an internet survey. *Social Science Research*, 36(1), 131-148; Schonlau M., Van Soest, A, Kapteyn, A., and Couper, M.(2009). Selection bias in Web surveys and the use of propensity scores. *Sociological Methods and Research*, 37(3), 291-318.

⁴² Link, M., (2013) Emerging Technologies: New Opportunities, Old Challenges. Presentation FedCASIC Workshop, Washington, DC, March 19, 2013

⁴³ Paul, M. and Dredze, M. *You Are What You Tweet: Analyzing Twitter for Public Health*. Michael J. Paul and Mark Dredze. Human Language Technology Center of Excellence. www.cs.jhu.edu/~mdredze/publications/twitter_health_icwsm_11.pdf

⁴⁴ http://unstats.un.org/unsd/statcom/statcom_2013/seminars/Big_Data/BigData_UNGlobalPulse_Kirkpatrick.pdf

correlation between privacy concerns and low engagement⁴⁵. The concerns mentioned in the survey can be grouped into concerns about identity theft in the financial world, concerns about unwanted spam, and solicitations in the digital world, as well as offline harm, stalkers, and employment risk in the physical world. A study of a selected group of college students found personality characteristics to be predictive of Facebook postings, with people who score high on the openness factor having less strict privacy settings, which may cause them to be susceptible to privacy attacks⁴⁶. The individual perception of risks involved with engaging in participation can probably be characterized in terms of cost-utility equations, just like participations in surveys⁴⁷. For many users the utility of connecting with friends, receiving and spreading information, searching and purchasing online outweighs the costs and risks, but it remains interesting to see if the recent privacy debates will change the equation for more people⁴⁸.

Overcoverage and Multiplicity

Overcoverage refers to the inclusion of units in the data that are not members of the target population, while multiplicity refers to the multiple appearances of a single person. On Facebook one can for example find pages for dead people or animals and errors in Geocodes⁴⁹ or disguised IP-addresses might erroneously identify users as being part of a geographically defined population of interest when they are not. Similarly, Paul and Dredze (2011) observed in their analysis of Twitter data that “some tweets, particularly those about more serious illnesses like cancer, were in regards to family members rather than the user. This could throw off geographic statistics if the subject of a tweet lives in a different location than the author of the tweet.”⁵⁰

⁴⁵ Staddon, J. Huffaker, D. Brown, I. and Sedley, A. Are privacy concerns a turn-off? Engagement and privacy in social networks. <https://static.googleusercontent.com/media/research.google.com/en/us/pubs/archive/38142.pdf>

⁴⁶ Halevi, T., Lewis, J., & Memon, N. (2013). Phishing, Personality Traits and Facebook. *arXiv preprint arXiv:1301.7643*.

⁴⁷ [Singer, Eleanor](#). 2011. "Toward a Benefit-Cost Theory of Survey Participation: Evidence, Further Tests, and Implications." *Journal of Official Statistics*, 27(2): 379-392.

⁴⁸ http://www.globalresearch.ca/nsa-spying-and-search-engine-tracking-technologies/5365435?utm_source=rss&utm_medium=rss&utm_campaign=nsa-spying-and-search-engine-tracking-technologies

⁴⁹ Zandbergen, P. A. (2009). Accuracy of iPhone locations: A comparison of assisted GPS, WiFi and cellular positioning. *Transactions in GIS*, 13(s1), 5-25.

⁵⁰ http://www.cs.jhu.edu/~mdredze/publications/twitter_health_icwsm_11.pdf

Multiplicity can occur for several reasons. Some people might open several Twitter accounts and Facebook pages, or use different credit cards to keep different aspects of their lives separated or to ease accounting. In simple scrapings of online data, those people would be included multiple times, again, which is no problem if a transaction is the unit of analysis but can be a problem if inference is made to a defined set of people. Depending on the research question, such de-duplication is a problem, most likely only be solvable through data linkage or other forms of additional identifying information, *creating again possible threats to privacy*.

Other Missing data

Aside from coverage, data can be missing for a variety of other reasons. Infrequent postings of Facebook users would be an example of missing data. In principle those people would be covered, but information on them is missing. Whether background information is missing because users choose to not reveal them, or because the data stream does not include these kinds of information, for analyses that try to explain behaviors or are interested in subgroup analyses, the lack of demographic and other background variables can be a big hindrance⁵¹.

In a presentation at the 2013 FedCasic Conference, Link reported that about one third of the demographic information is missing from Facebook users⁵². A study by Kosinski, Stillwell and Graepel tried to infer demographic information from digital records of behavior, Facebook Likes, for 58,000 Facebook volunteers. They classified African Americans and Caucasian Americans correctly in 95% of cases, and males and females in 93% of cases, “suggesting that patterns of online behavior as expressed by Likes significantly differ between those groups, allowing for nearly perfect classification⁵³.” Good prediction accuracy was achieved for more private information relationship status (65%), substance use (73%), and sexual orientation (88% for males, 75% for females).

⁵¹ Couper, M. P. (2013, December). Is the Sky Falling? New Technology, Changing Media, and the Future of Surveys. In *Survey Research Methods* (Vol. 7, No. 3, pp. 145-156).

⁵² Link, M., (2013) Emerging Technologies: New Opportunities, Old Challenges. Presentation FedCASIC Workshop, Washington, DC, March 19, 2013

⁵³ Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits and attributes are predictable from digital records of human behavior. *Proceedings of the National Academy of Sciences*, 110(15), 5802-5805.

While it is unknown how well the prediction models would perform for those that did not volunteer to participate in the study, these results raise important concerns about the possibility to keep information *purposefully private*. In the words of the authors: “the predictability of individual attributes from digital records of behavior may have considerable negative implications, because it can easily be applied to large numbers of people without obtaining their individual consent and without them noticing. Commercial companies, governmental institutions, or even one’s Facebook friends could use software to infer attributes such as intelligence, sexual orientation, or political views that an individual may not have intended to share. One can imagine situations in which such predictions, even if incorrect, could pose a threat to an individual’s well-being, freedom, or even life.”⁵⁴

Other forms of missing data are those in Big Data that appear on an event level. Here too privacy concerns can be the reasons. Consider for example the use of cell phone usage data to analyze traffic patterns. Users of cell phones carry their devices with them and thus leave a trace in the data files of everywhere they go. However, some users might turn the device off when they take trips that should not be known to anybody.

In his keynote lecture at the 2013 conference of the European Survey Research Association⁵⁵, Couper points out that behavior of people is likely to change based on *concerns about privacy*, and that this concern is likely to grow if the collection and use of Big Data become more broadly known. While some report behavioral changes in privacy and tracking settings⁵⁶ as result of concerns about data use, to our knowledge no data has been released by owners of major social networks sites on the effects of increased knowledge of data use. More research is needed here on how to address issues of consent to data capturing and data use (see discussion of Nissenbaum in

⁵⁴ Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits and attributes are predictable from digital records of human behavior. *Proceedings of the National Academy of Sciences*, 110(15), 5802-5805.

⁵⁵ Couper, M. P. (2013, December). Is the Sky Falling? New Technology, Changing Media, and the Future of Surveys. In *Survey Research Methods* (Vol. 7, No. 3, pp. 145-156).

⁵⁶ <http://www.fastcoexist.com/3015860/people-are-changing-their-internet-habits-now-that-they-know-the-nsa-is-watching>

this volume). Given that the data generation process spans several countries, an exchange with professional organizations across the globe might be useful (see Elias in this volume).

Solutions and New Challenges: Data Linkage and Information Integration

Many of the measurement and inference problems – whether they occurred because of privacy and confidentiality concerns or for other reasons – can be addressed through data linkage and other forms of information integration. Data linkage in particular has its own *privacy and confidentiality challenges*, but some of the other methods also have implications related to privacy and confidentiality.

In surveys, a variety of methods⁵⁷ that all fall under the heading of calibration can correct for coverage and nonresponse errors by using auxiliary data. Auxiliary data can be any information that is available for the target population, either for each individual population unit or in aggregate form. In a household survey, counts of persons in groups defined by age, race/ethnicity, and gender may be published from a census or from population projections that are treated as highly accurate⁵⁸. A necessary requirement for using those techniques is the *availability of comparable content and units of analysis* to link with information from more trusted sources. For example, if variables used for weighting are based on respondents' answers during the interview, then prior to data collection care should be taken that the questions asked in the survey match those of the benchmarking source. Even for demographic variables this seems like a straightforward task, but is often not that simple. The current U.S. Census, for example, allows for multiple-race categories, a feature many surveys or social network sites don't share. Non-probability surveys have started using such methods to attempt to make population estimates⁵⁹. For some Big Data sources this would require cooperation with companies to provide

⁵⁷ E.g. poststratification, general regression estimation, and raking.

⁵⁸ Valliant, Dever, Kreuter (2013). *Practical Tools for Sampling and Weighting*. Springer

⁵⁹ Valliant, R., and Dever, J. (2011), "Estimating Propensity Adjustments for Volunteer Web Surveys," *Sociological Methods and Research*, 40, 105-137 and Dever, J., Rafferty, A., and Valliant, R. (2008), "Internet Surveys: Can Statistical Adjustments Eliminate Coverage Bias?" *Survey Research Methods*, 2, 47-60. For a very accessible discussion of these approaches to nonprobability samples see Couper 2013 and AAPOR. (2013). *Report of the AAPOR Task Force on Non-Probability Sampling*". *Journal of Survey Statistics and Methodology*, 1, 90-143.

sufficient access to common variables, which can raise issues of confidentiality pledges given by the respective data collectors (see Stodden in this volume).

Another solution increasingly popular among surveys severely challenged by falling response rates, hard-to-reach populations, and stagnant survey budgets⁶⁰ is the use or linkage of administrative data to inform about coverage bias, nonresponse bias or measurement error in the survey data. Likewise, policymakers ask for evidence-based program evaluation, ideally with controlled randomized trials, but those often fail to be fully randomized when the practical data collection challenges kick in. And current movements towards comparative effectiveness mean increased interest in answering questions regarding effects outside the narrow study population⁶¹. Here too statisticians have started to combine multiple data sets or integrate information from different sources to mitigate the shortcomings of the original study, and high hopes are expressed with respect to organic Big Data to help here even more.

The act of combining several data sources, for example survey and administrative data, is not without its own challenges, and in several areas research is needed to ensure that this direction is one that can lead to success. Direct record linkage and statistical matching are two of the most common procedures for linking different data sources. Both methods have relative tradeoffs and rely on untested assumptions that threaten their practical use.

The quality of linked databases can be adversely affected by a) inconsistencies between information collected from respondents in the one data set (e.g. the survey) and information contained in the other (e.g. an administrative database or social network site information) and b) errors in the record linkage process itself⁶². Such errors can be due to erroneous linkage, imprecise matching variables, or the lack of respondent consent to link their survey responses to administrative data. It is often the case that the impact of these issues on the quality of the linked

⁶⁰ See the edited volume by Douglas S. Massey and Roger Tourangeau. The Nonresponse Challenge to Surveys and Statistics. The ANNALS of the American Academy of Political and Social January 2013; 645 (1)

⁶¹ Stuart, E.A., Cole, S.R., Bradshaw, C.P., and Leaf, P.J. (2011). The use of propensity scores to assess the generalizability of results from randomized trials. *The Journal of the Royal Statistical Society, Series A*. 174(2): 369-386. And Cole, S.R. and Stuart, E.A. (2010). Generalizing evidence from randomized clinical trials to target populations: the ACTG-320 trial. *American Journal of Epidemiology* 172: 107-115. [[Journal page](#)].

⁶² Smith, T. (2011). The report of the international workshop on using multi-level data from sample frames, auxiliary databases, paradata and related sources to detect and adjust for nonresponse bias in surveys. *International Journal of Public Opinion Research*, 23:389-402.

data sets is unknown. Evaluations of the procedures and criteria used in combining data sources and assessments of their impact on data quality are clearly needed if combined data are to become a key source of statistical information in the future.

Several studies have examined inconsistencies between observations made with survey and administrative data. A common assumption in those studies is that the records are error-free and suitable as a 'gold standard' against which survey responses can be compared, though this assumption often holds only for variables necessary to administer a certain program. Similar studies are needed for the various Big Data sources as well.

Prior to linkage, informed consent is usually needed to ensure that respondents are aware of the risks and benefits involved in releasing information for research purposes. Research has shown that consent rates to linkage requests vary widely from study to study, study population and country, and across the social, economic, and health fields, with percentages ranging anywhere from the mid-20's to the high-80's. But as in the discussion of coverage, here too it is not only the rate that matters but the difference between consenters and non-consenters. Studies have found gender, age, education, and wealth to be strongly related to the likelihood of consent; and health status, income item nonresponse, and prior-wave cooperation in longitudinal studies also correlate with linkage consent. Direct bias assessments with respect to a target variable, on the other hand, are rare⁶³.

We know by now that consent rates do not only vary by person characteristic and study topic but also as a function of the consent request itself. So far no studies have experimented with the stated risk of data linkage, but studies of informed consent for study participation have shown that stronger assurances of data confidentiality lead to higher response rates, with the exception that stronger assurances can backfire and lead to lower participation rates and more expressions of suspicion when the survey topic is innocuous and the confidentiality risk is minimal⁶⁴. Recent

⁶³ Sakshaug, Joseph; Kreuter, Frauke (2012): Assessing the magnitude of non-consent biases in linked survey and administrative data. In: Survey Research Methods, Vol. 6, No. 2, S. 113-122.

⁶⁴ Singer, E., Hippler, H. J., & Schwarz, N. (1992). Confidentiality assurances in surveys: Reassurance or threat? *International Journal of Public Opinion Research*, 4 (3): 256-268. , Bates, Dalhammer, Singer 2008. Privacy Concerns, Too Busy, or Just Not Interested: Using Doorstep Concerns to Predict Survey Nonresponse, *Journal of Official Statistics*, 24,4 591–612. Couper, Singer, Conrad, Groves (2010).

experimental studies on the wording of data linkage requests showed that framing of the request as well as the consent placement influences consent rates⁶⁵. The linkage indicator itself also matters, and more hesitations arise when social security numbers are used. For large scale linking, other identifiers that do not require self-reporting might be better suited. Geocodes of addresses might be the most fruitful approach,⁶⁶ though here too one can find differential error depending on the type of address used⁶⁷. Geography also allows the augmentation of other geographic information, which is a variable available in many Big Dats sources. If higher levels of aggregation are used, requirements on exact placement are less stringent.

In contrast to direct record linkage, statistical matching, or the act of matching “similar” records (in a statistical sense) based on a set of matching variables common to both data sets, does not require obtaining consent from survey respondents and can be performed on the entire sample. Several matching procedures have been proposed. However, many of these procedures depend on the assumption of conditional independence, asserting that any systematic differences between two or more data sources are explained by the matching variables. Only when this assumption is met will the true underlying distribution of all variables in both data sets be reflected in the matched data set; otherwise, statistical relationships estimated from the matched data set will be misleading.

One problem that cannot be solved by either technique is if people are missing for privacy reasons from all data sources that are intended for linkage. For example, nonrespondents in surveys might

Experimental Studies of Disclosure Risk, Disclosure Harm, Topic Sensitivity, and Survey Participation, *Journal of Official Statistics*, 26, 2, 287-300

⁶⁵ Sakshaug, J.; Tutz, V.; Kreuter, F. (2013): Placement, wording, and interviewers: Identifying correlates of consent to link survey and administrative data. In: *Survey Research Methods*, Vol. 7, No. 2, S. 133-144: Kreuter, F., Sakshaug, J. Tourangeau, R. Using Gain-Loss Framing to Obtain Respondent Consent to Link Survey and Administrative Data (under review)

⁶⁶ For an overview and new development see Schnell, R.: Combining Surveys with Non-Questionnaire Data: Overview and Introduction; in: Engel, U., Jann, B., Lynn, P., Scherpenzeel, A. & Sturgis, P. (ed.): *Improving Surveys Methods: Lessons from Recent Research*, Routledge, Taylor & Francis Group. in Print; and Schnell, R.: Getting Big Data but preventing Big Brother: Entwicklung neuer technischer Lösungen für die datenschutzgerechte Zusammenführung personenbezogener Daten; in: *UNIKATE 45: Fusionen - Universität Duisburg-Essen*. in Print

⁶⁷ Eckman, S. And English, N. Creating Housing Unit Frames from Address Databases Geocoding Precision and Net Coverage Rates *Field Methods* 24 (4), 399-408

also not be active on Facebook or decline the use of Credit Cards. To our knowledge there has so far not been any systematic research on this issue.

Another real challenge to data integration is the linkage of data with different privacy restrictions. Several chapters in this book provide suggestions, notably Karr and Reiter and Greenwood et al.. Infrastructure needs to be in place that allows linkage and the addition of identifiable IDs to place people in locations, de-duplicate, etc..

As the former Census Director Robert Groves summarized on the Director's Blog⁶⁸: "The challenge to the Census Bureau is to discover how to combine designed data with organic data, to produce resources with the most efficient information-to-data ratio. [...] Combining data sources to produce new information not contained in any single source is the future. I suspect that the biggest payoff will lie in new combinations of designed data and organic data, not in one type alone."

Discussion

Some of the challenges of Big Data may, in some sense, be addressed by adapting the approaches of "small data" to a larger scale. Designed Big Data experiments are already commonplace, particularly with so-called A/B testing in industry applications⁶⁹, and likely will see much wider application. We will almost certainly encounter methodological challenges as we continue to move into Big Data and so the corresponding research will be essential. Another development that may serve to simplify many Big Data problems is the continuous refinement and sharpening of scientific questions and hypotheses. While one of the key advantages of Big Data is that it allows one to explore and discover previously unobserved patterns, such free-form exploration can lead to numerous problems with respect to inference and measurement, some of which we have discussed in this chapter. Eventually, science will move beyond the fascination with terabytes and petabytes, and the ability to collect data, and focus more on their ability to analyze data and address

⁶⁸ Groves, R. "Designed Data" and "Organic Data", Director's Blog; <http://directorsblog.blogs.census.gov/2011/05/31/designed-data-and-organic-data/>

⁶⁹ Kohavi, R., Longbotham, R., Sommerfield, D. Henne, R.M. 2009. Controlled Experiments on the Web: survey and practical guide. Data Min Knowl Disc 18:140–181

specific questions. As investigators become more comfortable and familiar with the scope of Big Data, they will develop a much better understanding of what kinds of hypotheses can be address by such data and, more importantly, what kinds cannot.

Some research questions can only be addressed through the combination of data products. Thus research on linkage and matching needs to be expanded, covering the issues raised above and preparing for the analysis of linkage (consent) bias. When done successfully, linkage can provide a rich source, not just for substantive analysis when surveys, experiments, administrative data, sensor data or other organic Big Data are respectively enriched, but also to evaluate the quality of the linked sources, including research on the missing elements in both.

As the march of Big Data progresses, continuous effort will be needed to ensure standards of transparency in the collection of data, so that there may be appropriate accountability and so that valid statistical inferences can be made. Principles developed for traditional data collection efforts, such as the use of metadata and paradata to describe and monitor data-generating processes, will need to be translated to modern high-volume data collection methods such as sensor networks, real-time data streams, and imaging modalities. So far there is no context-independent framework for evaluating data-generating processes but it would more thinking in this area would be useful.

Transparency in the data collection goes hand in hand with transparency on the back end. The issues related to reproducibility of any scientific investigation are potentially magnified in a Big Data analysis. Naturally, any linkage, data preparation and processing, analysis, and reporting needs to be fully transparent so that the handling of these problems may be open to critique. However, because of the increased complexity of documenting actions taken to big data, we will tend to rely on computer code descriptions rather than more traditional written forms. Ultimately, ensuring reproducibility through the proper documentation of any statistical analysis will be critical because the uniqueness and size of Big Data studies will be less likely to be replicated in other investigations. It is in these cases that ensuring reproducibility of the analysis will be essential. (SEE STODDEN CHAPTER).

Because many datasets are too large to travel frequently from one place to the next, or privacy concerns may prevent them from being examined at an insecure location, data processing at the originator's site or through trusted 3rd parties⁷⁰ will be an essential element in working with these combined data products. Although examples of such procedures are already in place (e.g. National Center for Health Statistics Research Data Centers), more work is needed to create legal agreements to allow for data access, solve issues of liability, international border crossing, and mismatches in confidentiality standards. However, one potential upside of “moving computations to the data” is that such remote data processing will require people to document and share the code for their analyses. Hence, there may be a built-in reproducibility safeguard if datasets have to be analyzed remotely. It is these changes in data processing that will, if implemented in the right way (SEE a discussion by Landwehr in this volume), ultimately lead to more transparency and with that hopefully to more trust by individuals in the use of Big Data (at least in the context of statistical analyses), and to better data quality.

⁷⁰ http://unstats.un.org/unsd/statcom/statcom_2013/seminars/Big_Data/BigData_OECD_Wyckoff.pdf